Injong Won

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Education

Class 2025 **B.Sc. Computer Science & Statistics**, *University of Toronto*, Toronto, ON, Honours.
- Teaching Assistant: **CSC369 Operating Systems** | Graduate Coursework: **CSC413 Neural Networks & Deep Learning**

Audit **Deep Learning Systems**, *Carnegie Mellon University*.

Technical Skills

Languages TypeScript, Node.js, Python, C, C++, Rust, CUDA, Java, SQL

Infra & Tools Docker, Prometheus, Grafana, CI/CD, TensorRT-LLM, vLLM, PyTorch, FSDP, Linux

Professional Experience

2026 **Software Engineer,Training (Contractor)**, *LinkedIn*, Remote.
- Built **LLM evaluation harnesses** with automated scoring rubrics for code generation (Python, TypeScript, SQL); engineered adversarial test cases for correctness, edge-case robustness, and efficiency—outputs fed directly into LinkedIn's **fine-tuning** and **RLHF** pipeline, establishing **evals and regression tests** for agentic workflows.

2025 **Undergrduate Research**, *Vector Institute*, Toronto, ON.
- Architected end-to-end **ML pipeline observability** using Prometheus and Grafana for a platform serving **18k+ concurrent users**; established latency SLOs, automated regression detection, and built **real-time dashboards** reducing incident MTTR by **35%**.

2023 **Software Engineer**, *RBC Capital Markets*, Montreal, QC.
- Profiled and optimized latency-sensitive **data pipelines** in trading risk/pricing infrastructure; reworked **caching layers** and fetch paths on P&L-sensitive flows, achieving **70% end-to-end latency reduction**—directly applicable to low-latency **streaming output** and **IPC** architecture.

2019–2020 **Software Engineer**, *IBM Watson AI*, Markham, ON.
- Contributed to **Watson Commerce Insights**, an enterprise AI/ML analytics platform; built CI/CD test infrastructure and integration suites reducing production regressions across **ML-powered** release cycles.

Selected Projects

2025 **Foundation Model Training Pipeline**, *PyTorch · Triton · FSDP · NCCL · CUDA · Python*.
- Built e2e **pretraining pipeline** from raw Common Crawl (**BPE tokenizer** from scratch, quality filtering, deduplication) into a pre-norm Transformer (**RoPE, SwiGLU, GQA**) trained with AdamW + cosine LR; matched **GPT-2 small perplexity within 1.8%** at compute budget via **Chinchilla scaling**.
- Implemented **FlashAttention-2** from scratch in Triton (**2.1**× over PyTorch on A100); scaled to **4×A100s** with **FSDP** ZeRO-3, **NCCL** all-reduce, gradient checkpointing, BF16—**3.2**× throughput, **41%** peak memory reduction vs. single-GPU baseline.

2025 **LLM Inference Serving & Benchmarking**, *TensorRT-LLM · vLLM · sglang · Triton · CUDA*.
- Benchmarked **TensorRT-LLM** (INT8 quantization, in-flight batching) vs. **vLLM** (PagedAttention, continuous batching) across batch sizes 1–64 on A100; applied **FP8** to attention layers; characterized **latency vs. throughput tradeoffs** across serving configurations to identify optimal deployment settings for different workload profiles.

2025 **Reinforcement Learning for LLM Alignment**, *Pytorch · RLHF*.
- Implemented **PPO-style RLHF loop**: policy rollout generation, per-token **KL-penalty** against frozen reference model (β-ablation), **GAE advantage estimation** (λ sweep), and clipped surrogate objective—validated alignment improvement via held-out reward scores.

2024 **GPU Kernel Optimization**, *C++ · CUDA · Triton · Nsight Compute · Python*.
- **CUDA Matmul:** Tiled shared-memory kernel reaching **14.2 TFLOP/s** (87% of cuBLAS peak), **823 GB/s** bandwidth—**3.1**× over naive; **Nsight Compute** roofline analysis guided register tiling, warp occupancy, and memory coalescing optimizations.